

Master Math for JEE Main & JEE Advanced

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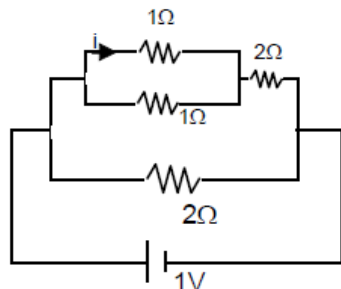
JEE Mains 2020 Jan Chapter wise Question Bank

Current Electricity

Q1

The current 'i' in the given circuit is

दिये गये परिपथ में धारा 'i' है –



(1) 0.2A

(2) 0.3A

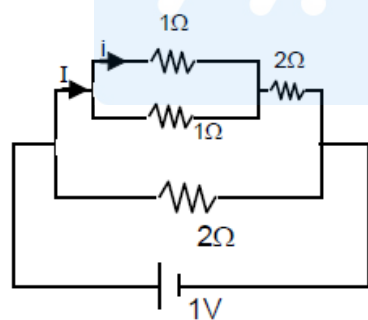
(3) 0.5A

(4) 0.25A

7th Jan Morning

Sol

(1)



$$I = \frac{1}{2.5} = 0.4A$$

$$i = \frac{I}{2} = 0.2A$$

Q2

In a house 15 Bulbs of 45 W, 15 bulbs of 100 W, 15 bulbs of 10 W and Two heaters of 1 KW each is connected to 220 V Mains supply then find minimum fuse current

एक घर में 45W के 15 बल्ब 100W के 15 बल्ब, 10W के 15 बल्ब तथा 1KW के दो हीटर 220V की मुख्य धारा से जुड़े हुए हैं, तो फ्यूज तार से प्रवाहित न्यूनतम धारा का मान होगा।

(1) 5 A

(2) 20 A

(3) 25 A

(4) 15 A

Sol

(2)

Total power is कुल शक्ति $(15 \times 45) + (15 \times 100) + (15 \times 10) + (2 \times 1000)$
 $= 4325 \text{ W}$

So current is अतः धारा $= \frac{4325}{220} = 19.66 \text{ A}$

Ans is 20 Amp.

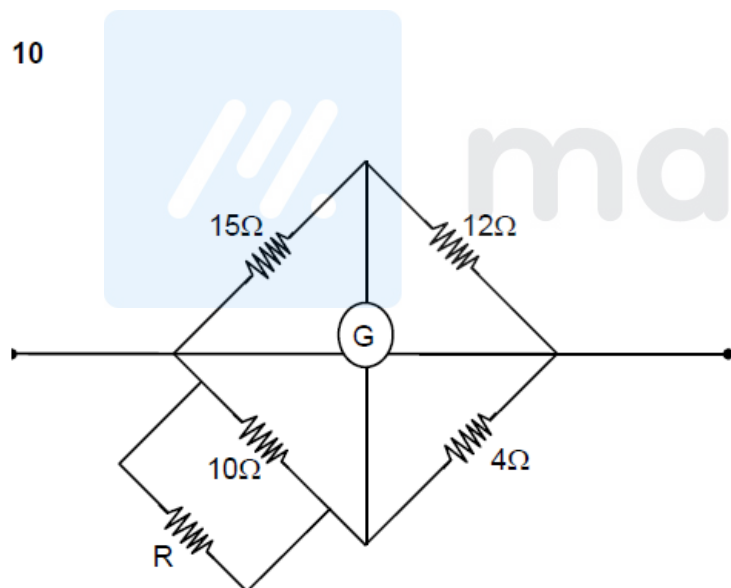
Q3

Four resistors of 15Ω , 12Ω , 4Ω and 10Ω are arranged in cyclic order to form a wheatstone bridge. What resistance (in Ω) should be connected in parallel across the 10Ω resistor to balance the wheatstone bridge.

8th Jan Morning

Sol

10



$$\frac{10R}{10+R} \times 12 = 15 \times 4$$

on solving

$$R = 10\Omega$$

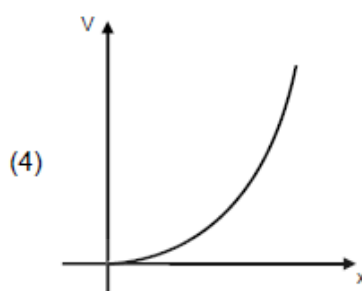
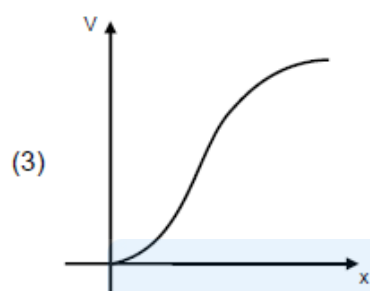
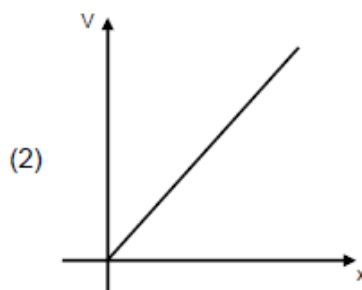
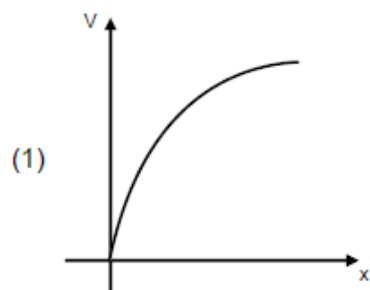
Q4

Current Electricity

JEE Mains 2020 Jan Chapter wise Question Bank

A charge particle of mass m and charge q is released from rest in uniform electric field. Its graph between velocity (v) and distance travelled (x) will be :

एक समान वैद्युत क्षेत्र में एक आवेश q जिसका द्रव्यमान m है छोड़ा गया है। इसके द्वारा चली गयी दूरी (x) और वेग (v) के बीच ग्राफ होगा।

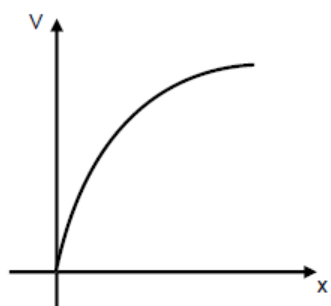


8th Jan Evening

Sol

(1)

$$v^2 = \frac{2qE}{m}x$$



Q5

Two batteries (connected in series) of same emf 10 V of internal resistances 20Ω and 5Ω are connected to a load resistance of 30Ω . Now an unknown resistance x is connected in parallel to the load resistance. Find value of x so that potential drop of battery having internal resistance 20Ω becomes zero.

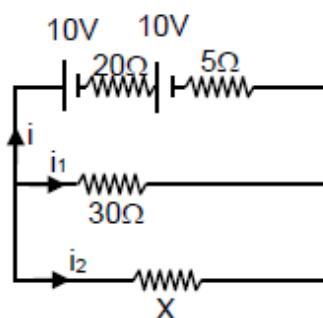
Sol

30

$$V_1 = \varepsilon_1 - i \cdot r_1$$

$$0 = 10 - i \times 20$$

$$i = 0.5\text{A}$$



$$V_2 = \varepsilon_2 - i r_2$$

$$= 10 - 0.5 \times 5$$

$$V_2 = 7.5\text{V}$$

$$0.5 = \frac{7.5}{30} + \frac{7.5}{x}$$

$$0.5 = 0.25 + \frac{7.5}{x}$$

$$\frac{7.5}{x} = 0.25$$

$$x = \frac{7.5}{0.25} = 30$$

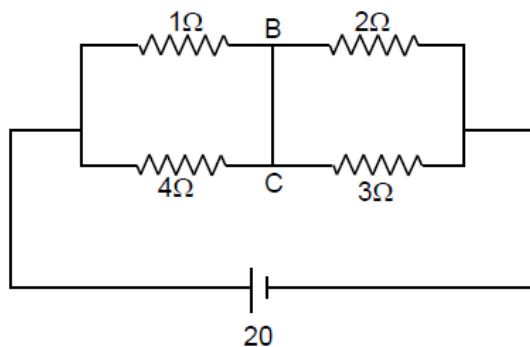
Q6

Current Electricity

JEE Mains 2020 Jan Chapter wise Question Bank

Find current in the wire BC.

तार BC में धारा ज्ञात करो—



(1) 1.6A

(2) 2A

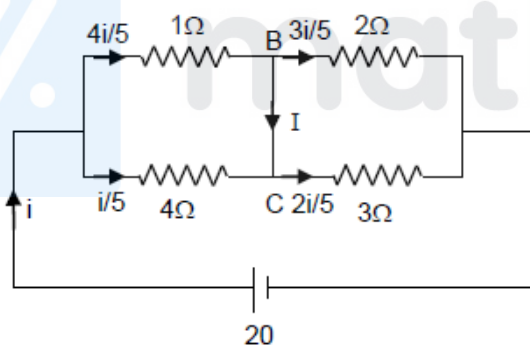
(3) 2.4A

(4) 3A

9th Jan Morning

Sol

(2)



$$R_{\text{eff}} = \frac{4}{5} + \frac{6}{5} = 2\Omega$$

$$i = \frac{20}{2}$$

$$= 10\text{A}$$

$$I = \frac{4i}{5} - \frac{3i}{5} = +\frac{i}{5} = 2\text{A}$$

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